

winding companies to compete with multinational companies. Reliance on import for raw materials, specifically copper, is a major handicap for Indian transformer manufacturers in achieving faster turnaround time as well as lower production costs.

Major components are imported from China and the power section is designed for ideal conditions. This is the reason for a lot of failures across the power supply section, as it is designed for ideal conditions; whereas in India, power distribution conditions are not so ideal, believes Bharara. He says, "This, in turn, has been a major boost for Indian coil winding products, even though power supplies integrated design comes from outside India, which simultaneously poses as a threat. He adds, "At the present moment, the primary competitor is China because they have the power of practical authority."

Another problematic area is transportation. "We usually receive a shipment from China in about 17 days from the date of order, but transporting a shipment from Mumbai to Delhi takes about 20 days," says Bharara. The reason for China's massive growth is a clearly-defined core infrastructure. He adds, "One of the reasons why China is ahead in the game is because co-operative banks charge very nominal interest. Therefore Chinese manufacturers can afford to not include the interest expenditure to the cost of the product."

Finance is another area that needs attention from the government. Bharara says, "Loans are granted at 13 per cent to 14 per cent interest, which makes it very difficult for an Indian manufacturer to survive."

Material and machinery problems

Most of the raw material for making coil wound components is imported from China. Iron oxide is, however, naturally available in India. Although ferrite forms the core for most wound

Market trends in coil components

- Water-cooled technology based components
- Preference for dry-type resin-cast transformers rather than oil-cooled
- Preference for amorphous-core rather than traditional silicon steel

—Contributed by Navin Upreti, vice president-operations, Tamara Elcomponics Tech Pvt Ltd

components, ferrite industry does not attract many workers due to the working conditions and hazards involved, making it dependent on mechanisation. Mechanisation is capital-intensive and requires very large volumes to make it feasible in the long run.

National Physical Laboratory, New Delhi, had developed indigenous technologies and given it to 13 different companies, four of which implemented those but all of them discontinued due to one reason or the other. Proven technologies from other countries were relied upon. Upgrading or modifying factories was an issue. As a result, Indian ferrite manufacturers produced lower quality products at low or medium volumes. Today, ferrite cores are imported by a lot of companies, even from Germany.

Winding machinery used is mostly of Chinese make. German machinery is very expensive. Repairs are seldom required but are extremely expensive. Indian machines are of similar quality and maintenance as their Chinese counterparts, but lose out to them marginally due to price.

Man-management has also been a major problem considering this industry has enjoyed cheap labour until recently. Work is quite mundane. Quality checks are done manually in different stages. Increasing operational efficiency is an area that could make a difference.

There is hardly any quality affiliation in this field. Identifying companies and placing orders are mostly done through personal contacts.

Opportunities that need to be capitalised on

India has a huge consumption demand. According to ELCOMOS report

(by ELCINA), the demand for electronics in the country is expected to reach US\$ 400 billion by 2020. This growing market provides a huge opportunity for the components sector, but manufacturing in India needs a boost. Manufacturing of end-use products happens largely through original equipment manufacturers (OEMs) and electronic manufacturing services (EMS) companies. Local design know-how and manufacturing technology skills boost opportunities for coil wound components manufacturing in India. "Printed circuit board (PCB), liquid crystal display (LCD) and light emitting diode (LED) industries need a lot of capital expenditure (CAPEX), whereas for the coil wound components industry, there is not much CAPEX required in terms of equipment," informs Bharara.

In the coil wound industry, raw material prices are more or less similar for companies everywhere. "We can definitely compete with China and offer better pricing, since we are buying raw materials at international prices because wages of labourers in China have increased," informs Bharara. Although the output of Indian labour is half in comparison to China, the rise in wages in China might prove to be a breather. He adds, "This way this particular segment can perform very well." Production costs can be reduced with better management techniques.

Wound components are used in almost all electronic and electrical products; hence, the market is undoubtedly going to rise.

Key sectors for manufacturers to look out for

New technologies are changing the geometry size of the coil wound industry. Smart City concept is an